

Course Evaluation

The course evaluation is available in Blackboard.

Please complete online course evaluation at Blackboard, which closes in April 27

Your feedback is crucial to improving the course

Final Exam Schedule

- Final exam has been scheduled:
- 12:30 p.m. – 3:00 p.m., Tuesday, May 5
- Requirement:
- Closed book and closed notes
- One-page double-sided cheat sheet is allowed
- A calculator is allowed for $+ - * /$
- **It will cover all the topics discussed in class**

Review of Chapter 2- Chapter 5

Chapter 2

- Human vision system
- Basics of image processing

Chapter 3

- Intensity transformation
- Spatial filtering

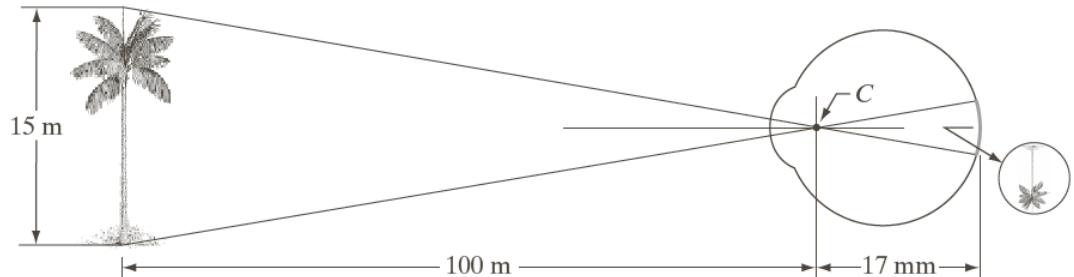
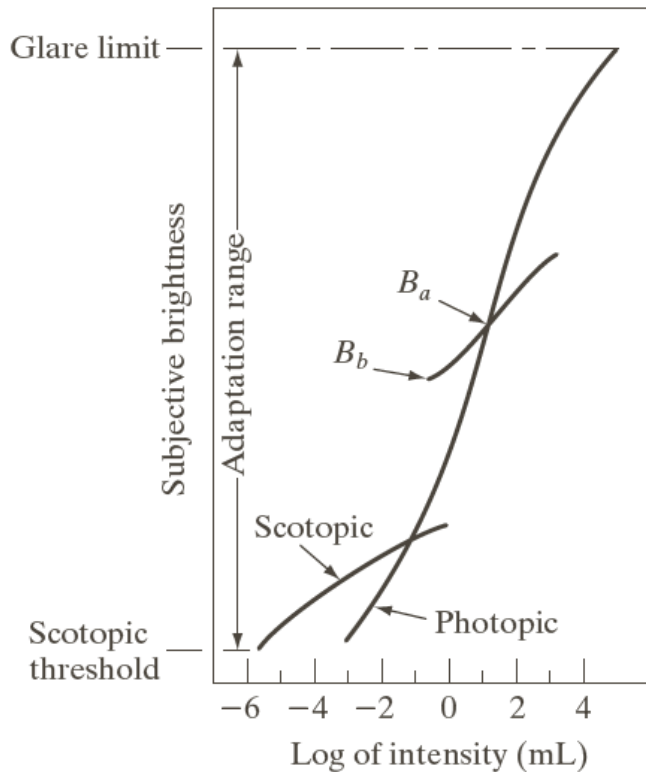
Chapter 4

- Fourier transform
- Image convolution in frequency domain

Chapter 5

- Image denoise
- Image degradation
- Image restoration

Human Vision System



- **Geometrical relationship between the real object and the image of the object**
- The minimum size of the object you can “see”

Brightness adaption

Basics of Image Processing

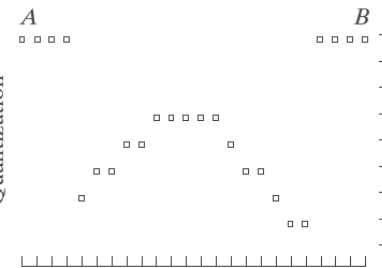
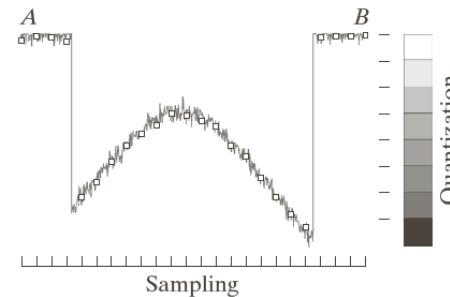
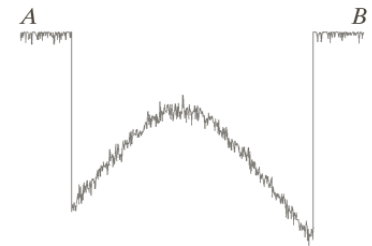
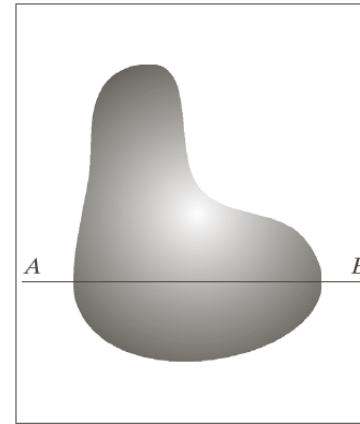
$$f(x, y) = i(x, y) \cdot r(x, y)$$

- **Image sampling and quantization**
 - Spatial/intensity resolution

- **Dynamic range of the image**

$$I_{\max} / I_{\min}$$

- **Image representation and storing**
- **Image interpolation**
 - Nearest neighbor and bilinear
- **Set operations**



Basics of Image Processing (Cont.)

Basic relationships between pixels

- Adjacency
- Connectivity
- Path

Basic relationships between regions

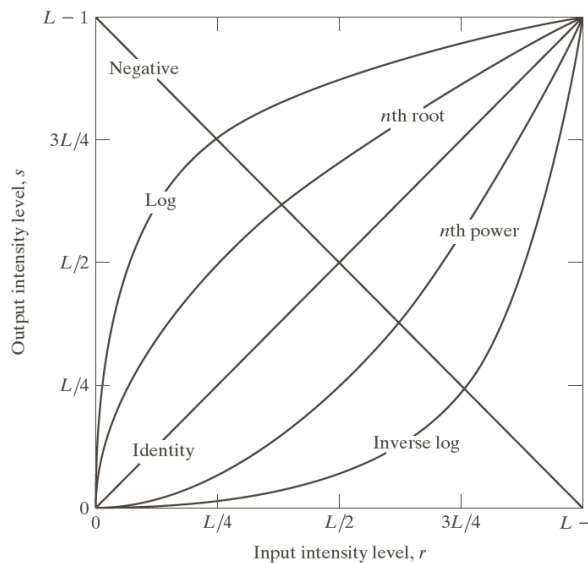
- Adjacency
- boundary

Distance measurement

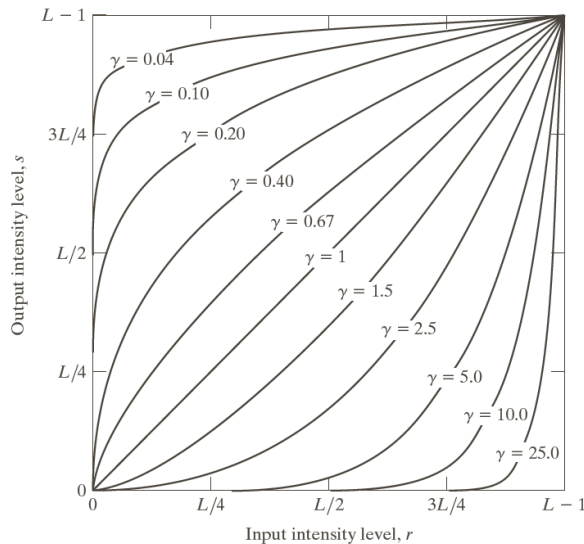
Mathematical tools

- Difference between matrix and array operation
- Linear/nonlinear operation
- Applications of image averaging, subtraction, and multiplication

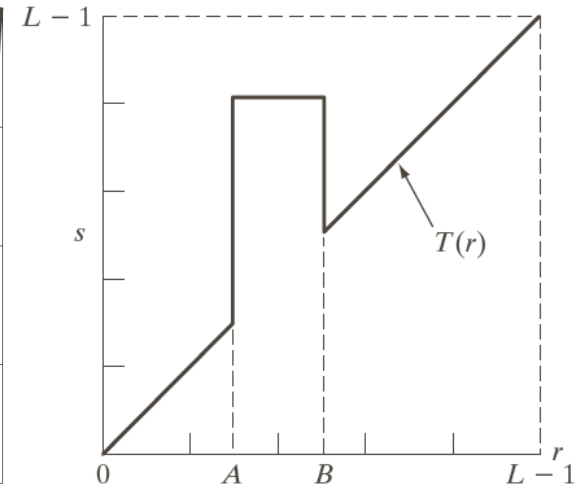
Intensity Transformation



Log transformation



Power-law
(gamma)
transformation



Intensity level
slicing

Applications and working conditions using these transformations

Histogram Processing

What is a histogram of an image?

Histogram equalization

Histogram matching

Spatial Filtering

Image convolution in spatial domain and has properties of

- Commutativity, Associativity, distributivity

Image correlation in spatial domain

Spatial filters

- Smoothing filter
 - Average filter
- Sharpening filter
 - Laplacian filter
 - Unsharp masking
 - Sobel operator
- Order-statistic filter
 - Median filter
 - Min/max filter

Given a filter, you should know what it is use for

You should know what filter you can use for a specific task

Fourier Transform

Fourier series

$$f(t) = \sum_{n=-\infty}^{+\infty} c_n e^{\frac{j2\pi nt}{T}}$$

Unit impulse and its sifting property

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0)$$

Fourier transform

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt \quad \longleftrightarrow \quad f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

Image convolution in frequency domain

Basic Properties of FT

Linearity $h(t) = af(t) + bg(t) \leftrightarrow H(u) = aF(u) + bG(u)$

Translation $h(t) = f(t - t_0) \leftrightarrow H(u) = e^{-j2\pi t_0 u} F(u)$

Modulation $h(t) = e^{j2\pi u_0 t} f(t) \leftrightarrow H(u) = F(u - u_0)$

Scaling $h(t) = f(at) \leftrightarrow H(u) = \frac{1}{|a|} F\left(\frac{u}{a}\right)$

Conjugation $h(t) = f^*(t) \leftrightarrow H(u) = F^*(-u)$

Symmetry $f(t) \leftrightarrow F(\mu) \Rightarrow F(t) \leftrightarrow f(-u)$

Image Degradation

$$g(x, y) = h(x, y) \otimes f(x, y) + \eta(x, y)$$

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

Important noise models

- Gaussian noise model
- Impulse noise model

Image denoise

- Various mean filters and their applications
- Order-statistic filters and their applications

Image restoration

- Inverse filtering, Wiener filtering, and Constrained Least Square filtering
 - Working conditions

Fundamentals of color image processing

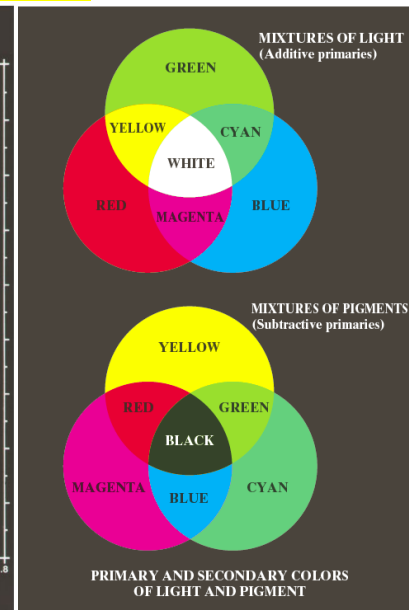
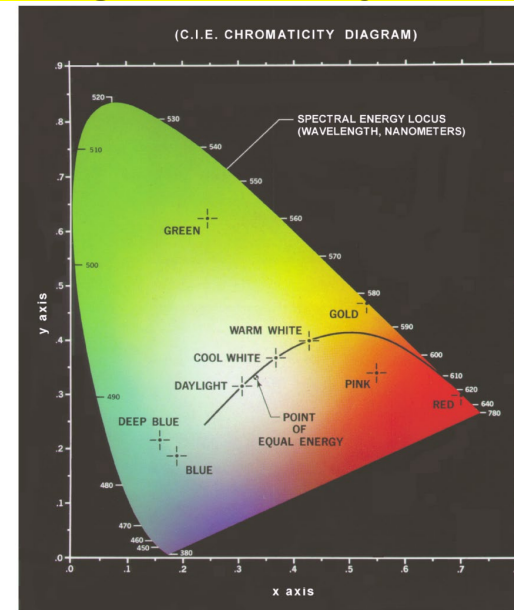
Primary/secondary colors

Primary/secondary pigments

Chromaticity: hue and saturation

Color gamut: any color on a line segment can be generated by two ending points; the same color can be generated by different combinations

What's the difference between them?

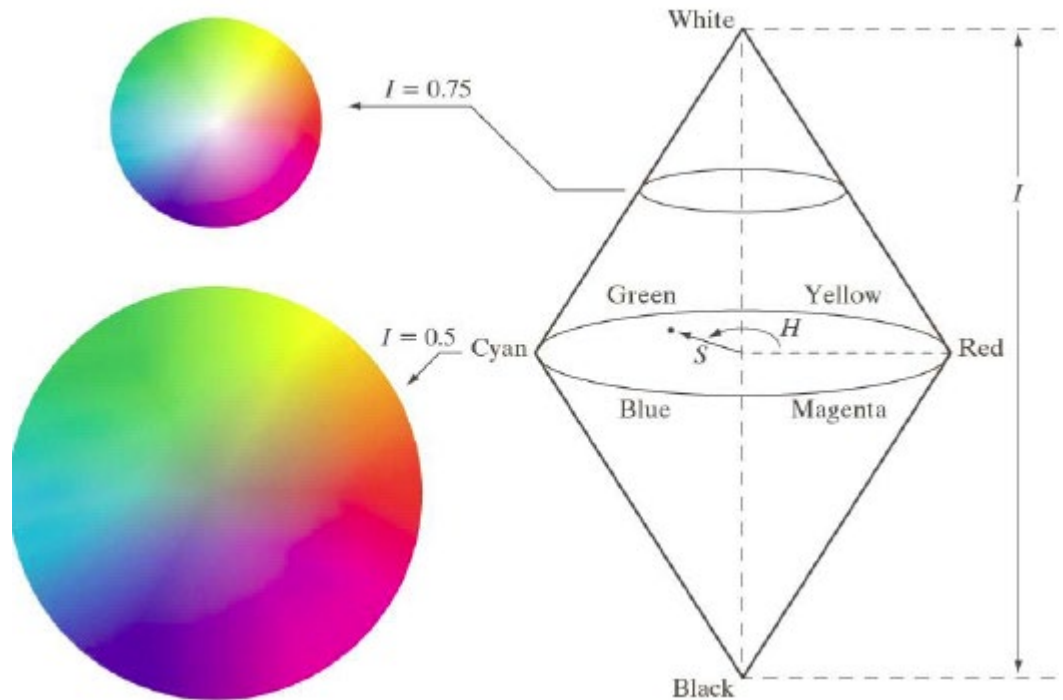


Fundamentals of color image processing (Cont'd)

RGB model

CMYK model

HSI model

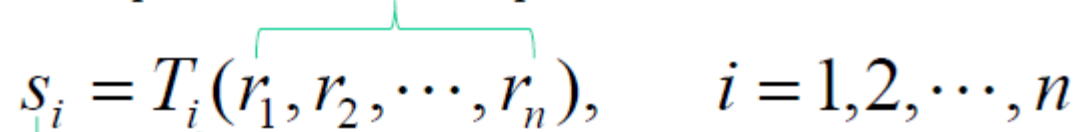


Requirement: how to represent a color in a specific model?

Color Transformation

For a color image with n components

input values for all components

$$s_i = T_i(r_1, r_2, \dots, r_n), \quad i = 1, 2, \dots, n$$


Output value for i th component Transformation functions

Intensity modification

HSI model

Color complement

RGB model

Tonal correction

HSI model

Color balancing

The choice of color model varies for a specific image

Histogram processing

HSI model

Which model is the most effective to perform a specific transformation?

Fundamentals of Image Compression

- Compression ratio $C = \frac{b}{b'}$
- Data redundancy $R = 1 - \frac{1}{C}$
- Three types of data redundancy and their examples

- Coding redundancy
- Spatial/temporal redundancy
- Irrelevant information

- Entropy and average number of bits $H = -\sum_{k=0}^{L-1} p_r(r_k) \log_2 p_r(r_k)$

- Objective fidelity criteria $L_{\text{avg}} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k)$

$$e_{\text{rms}} = \left[\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2 \right]^{\frac{1}{2}}$$

$$SNR_{\text{ms}} = \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \hat{f}(x, y)^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2}$$

Image Compression Methods

- Huffman coding
- Arithmetic coding
- Run-length coding

Requirement: encode/decode with these methods, working conditions for these methods; develop a compression method with a combination of these methods for a specific task

Edge-based Image Segmentation

- Connectivity
- Edge detectors
 - First-order derivative: Roberts (2x2), Prewitt (3x3), Sobel (3x3)
 - Second-order derivative: Laplacian
 - Complex edge detectors: LoG, DoG, and canny

Requirement: characteristics of these edge detector, perform simple edge detectors (1st and 2nd order) on an image

- Edge linking
 - Hough transform

Requirement: find a basic shape (e.g. a line) using Hough transform

Brief Review on Simple Edge Detectors

- First-order derivative
 - Roberts (2x2)
 - Prewitt (3x3)
 - Sobel (3x3, smooth + difference)
 - Thicker edge
 - One operator for one edge direction
- Second-order derivative
 - Laplacian (3x3)
 - Double edge
 - Zero-crossing
- Common issues:
 - Sensitive to image noise
 - Cannot deal with the scale change of the image

Brief Review on Advanced Edge Detectors

- **Laplacian of Gaussian (LoG)** $\nabla^2 G(x, y)$
 - Gaussian smoothing + Laplacian
 - Localize the edge position based on zero-crossings
 - Spaghetti effect of zero-crossings
 - Multiple filtering processing for scale variations
- **Difference of Gaussian (DoG)**
 - Approximation of LoG
 - Two different sigmas handle the scale variations
- Canny edge detector
 - Gaussian smoothing + 1st order derivative
 - Nonmaxima suppression to get single edge
 - Double thresholding and connectivity analysis to detect and link edges
- Thresholding
 - **Key factors affect thresholding**
 - Separation between peaks
 - Noise level
 - Relative size of objects and background
 - Illumination and reflectance

Region-based Image Segmentation

- Region growing
- Region-splitting and merging

Requirement: develop a region-based image segmentation method for a specific image

Basics of Mathematical Morphology

- Set reflection and translation $\hat{B} = \{w \mid w = -b, b \in B\}$ $(B)_z = \{c \mid c = b + z, b \in B\}$
- Erosion and dilation
- Opening: $A \circ B = (A \ominus B) \oplus B$
 - Smooth the contour of an object: smoothing outer corners
 - Break narrow isthmuses
 - Eliminate thin protrusions
- Closing: $A \bullet B = (A \oplus B) \ominus B$
 - Fill narrow breaks and gaps
 - Eliminate long and thin gulfs
 - Eliminate small holes
- Hit-or-miss transform
- Other operations
 - Thinning and thickening
 - Hole filling
 - Connected component analysis
 - Convex hull
 - Skeleton

Requirement: the applications of these operations; perform erosion, dilation, opening, and closing operation on an image

Properties of Erosion and Dilation

- **Dilation is commutative** $A \oplus B = B \oplus A$
- **Dilation is associative** $A \oplus B \oplus C = A \oplus (B \oplus C)$
- **Dilation** $A \oplus (B \cup C) = (A \oplus B) \cup (A \oplus C)$
- **Erosion** $(A \cap B) \ominus C = (A \ominus C) \cap (B \ominus C)$
- **Erosion and dilation are duals of each other**

$$A \oplus B = (A^c \ominus \hat{B})^c$$

- $A \subseteq (C \ominus B)$ **if and only if** $(A \oplus B) \subseteq C$
- **If** $A \subseteq C$, $A \oplus B \subseteq C \oplus B$ **and** $A \ominus B \subseteq C \ominus B$

Properties of Opening and Closing

- Opening: $(A \circ B) \circ B = A \circ B$
 $(A \circ B) \subseteq A$

 $if\ A \subseteq C, A \circ B \subseteq C \circ B$
- Closing $(A \bullet B) \bullet B = A \bullet B$
 $A \subseteq (A \bullet B)$

 $if\ A \subseteq C, A \bullet B \subseteq C \bullet B$

Good luck on your final exam!